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BACKTESTING VAR AND EXPECTED SHORTFALL FOR THE S&P 500

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INTRODUCTION

The question of how much capital a financial institution needs to hold against potential trading losses depend on the accuracy of a risk model. The cost of getting it wrong was made painfully clear during the Global Financial Crisis of 2008. At that time, models calibrated to a long period of calm markets dramatically underestimated the risk of portfolios that subsequently suffered catastrophic losses. More than a decade later, the same basic challenge arose again during the COVID-19 crash of March 2020, when market volatility rose to levels that constant volatility models could not anticipate. What these episodes share is a common failure mode: a mismatch between the volatility environment used to calibrate the model and the volatility environment in which it is asked to operate. Dynamic models such as GARCH address this directly by allowing volatility to evolve over time and can be re-estimated recursively over rolling windows so that parameter estimates continuously adapt to changing market conditions.

Risk models are used to forecast the distribution of future returns, which is usually represented by two common risk measures: **Value at Risk (VaR)** and **Expected Shortfall (ES)**. VaR is defined as the maximum loss not exceeded over a given horizon at a stated probability and has been the dominant regulatory capital metric since its adoption under Basel II. Despite its practical appeal, VaR has a fundamental weakness: it says nothing about the magnitude of losses beyond the threshold it identifies. **Expected Shortfall (ES)**, defined as the expected loss conditional on exceeding VaR, addresses this weakness directly. The Basel Committee's Fundamental Review of the Trading Book (BCBS, 2019) formalized this view by replacing 99% VaR with 97.5% ES as the primary internal-models capital metric under Basel III. A model must now get both the frequency and the average severity of extreme losses right - and the two can diverge substantially depending on the distributional assumption used.

A key input to calculate both risk measures is the forecast for variance of returns. Equity returns are heteroskedastic: volatility clusters and spikes during crises and recedes during calm periods in ways that are partially predictable. This thesis compares two approaches to capturing this structure: **Exponentially Weighted Moving Average (EWMA)** and **Generalized AutoRegressive Conditional Heteroskedasticity (GARCH)** model. **EWMA** model, introduced by J.P. Morgan/Reuters (1996) within RiskMetrics framework, uses a geometrically declining weighted average of past squared returns. Following RiskMetrics the decay parameter is usually calibrated at $\lambda = 0.94$. The **GARCH (1,1)** model of Bollerslev (1986) estimates a mean-reverting conditional variance by maximum likelihood, allowing the data to determine the speed of both shock response to market data and the pace of reversion to equilibrium variance. Both are widely used in practice; which is better for risk measurement is an empirical question.

The central research question of this thesis is: which combination of volatility model and distributional assumption produces the most statistically adequate VaR and ES estimates for the S&P 500 index, as evaluated by formal backtesting at the 95% and 99% confidence levels?

The empirical analysis is based on approximately 1,000 weekly log-returns from January 2007 to March 2026, covering the GFC, the COVID-19 crash, and the interest rate shock of 2022 - three qualitatively distinct stress episodes that allow model performance to be assessed across very different market environments. Backtesting uses four complementary procedures: the Kupiec (1995) unconditional coverage test, the Christoffersen (1998) independence test, the Christoffersen and Pelletier (2004) duration test, and the McNeil and Frey (2000) Expected Shortfall test.

Three hypotheses guide the analysis. First, dynamic models will substantially outperform unconditional models during extended calm periods because they adapt to the current volatility regime. Second, the t-Student distributional assumption will materially improve ES estimates at 99% because S&P 500 weekly returns exhibit excess kurtosis of 7.94, a property that the Normal tail cannot adequately capture. Third, GARCH (1,1) with t-Student innovations will be the best-performing specification overall, because its mean-reverting variance structure and fat-tailed innovations together address the two main limitations of the simpler EWMA model.

The main contribution of this thesis is a systematic, side-by-side comparison of EWMA and GARCH under a unified backtesting framework applied to a long, crisis-rich sample. The inclusion in the sample the 2022 interest rate shock, i.e. a sustained and gradual bear market qualitatively different from the sudden crashes of 2008 and 2020, reveals an important and previously under-documented distinction between the two models in their response to different types of market stress.

The thesis is organised as follows: Chapter 1 establishes the theoretical framework, Chapter 2 presents the methodology, Chapter 3 reports the empirical results, a Summary concludes.

I. MARKET RISK MEASURES

This chapter lays the theoretical groundwork for the empirical analysis that follows in next chapters. It begins with Value at Risk - the risk measure that has shaped regulatory practice since the 1990s - and then turns to Expected Shortfall, which has superseded it under the Basel III framework. After comparing these two measures directly, the chapter surveys the key empirical regularities of equity return volatility that motivate the modelling choices made in Chapter 2.

1.1 Value at Risk: Definition and Properties

Value at Risk entered mainstream risk management through J.P. Morgan's RiskMetrics initiative in 1994 and was codified as the standard capital metric under the Basel II Accord. Its appeal was transparent: regardless of the complexity of an institution's portfolio, VaR reduced its risk to a single number: the loss that will not be exceeded on more than $(1 - \alpha)$ of periods. Formally, for a portfolio with return r over a given holding period:

$$VaR_{\alpha} = -F^{-1}(1 - \alpha)$$

where $F^{-1}(\cdot)$ is the quantile function of return distribution, i.e. $F(\cdot)$ is the cumulative distribution function of returns. Under Basel II, banks were required to compute the 10-day, 99% VaR using at least one year of historical data and to hold regulatory capital proportional to this estimate (Jorion, 2007). The practical virtues of VaR - compactness, ease of estimation, and aggregability across business lines - explain its rapid adoption.

These virtues, however, come at a significant theoretical cost. Three limitations are particularly consequential for the analysis in this thesis:

The first one is **insensitivity to tail severity**: VaR identifies the edge of the loss distribution but says nothing about the magnitude of losses once the threshold is crossed. Two portfolios can have identical VaR estimates while one suffers losses just marginally above the threshold and the other experiences catastrophic drawdowns. This asymmetry of information is directly relevant during crises, when the depth of the tail - not merely its location - determines whether an institution survives.

The second limitation is **non-subadditivity**. A coherent risk measure, in the sense of Artzner et al. (1999), satisfies the requirement that the risk of a combined portfolio never exceeds the sum of its parts — formally, $\rho(A + B) \leq \rho(A) + \rho(B)$ for any two portfolios A and B — which is a formal expression of the diversification principle. VaR does not satisfy this requirement. Consequently, an institution can sometimes reduce its reported VaR by concentrating risk in a single position rather than diversifying. This creates a perverse incentive structure in capital regulation: the very measure intended to constrain risk-taking can be gamed by increasing tail concentration.

The third limitation is **blindness to the tail beyond the threshold**. During the GFC, many institutions held portfolios whose VaR estimates appeared reassuring, while their actual tail exposures were catastrophic. The models were not necessarily wrong about the *frequency* of violations; they were structurally incapable of communicating the *magnitude* of losses in the region that mattered. This is the failure mode that motivated the regulatory shift to Expected Shortfall.

1.2 Expected Shortfall as a Superior Risk Measure

Expected Shortfall (ES), which is also called Conditional Value at Risk (CVaR) or Expected Tail Loss, was proposed by Artzner et al. (1999) as a theoretically rigorous alternative that directly addresses VaR's deficiencies. At confidence level α, ES is defined as:

$$ES_\alpha = -E\,[\,r \mid r < -VaR_\alpha]$$

Rather than locating the edge of the loss distribution, ES measures the average loss across all outcomes in the VaR region: conditional on exceeding VaR, how large is the loss on average? This is the question whose answer determines the adequacy of capital buffers in a genuine crisis.

The theoretical superiority of ES over VaR rests on its status as a **coherent risk measure** in the sense of Artzner et al. (1999). It satisfies monotonicity, translation invariance, positive homogeneity, and - most importantly - **subadditivity**: the risk of a diversified portfolio never exceeds the sum of its components' risks. This eliminates the gaming possibilities that plague VaR and makes ES consistent with the economic logic of portfolio theory.

The regulatory transition was formalised in the Basel Committee's Fundamental Review of the Trading Book (BCBS, 2019), which replaced 99% VaR with 97.5% ES as the primary metric for banks using internal models. The motivation was explicit: 99% VaR had failed to capture the magnitude of tail losses during 2008, and ES - by averaging over the entire dangerous region rather than merely identifying its boundary - provides a more economically meaningful basis for capital regulation. This regulatory shift is the direct motivation for evaluating models on their ES performance alongside VaR in Chapter 3.

Property	Value at Risk	Expected Shortfall
Definition	Quantile of the loss distribution	Expected loss beyond the VaR threshold
Coherence	Not coherent — not subadditive	Coherent (Artzner et al., 1999)
Tail sensitivity	None ignores losses beyond threshold	Full averages all tail losses
Regulatory use	Basel I–II primary metric (99% VaR)	Basel III / FRTB standard (97.5% ES)
Backtesting	Frequency of violations only	Magnitude of violations required
Gaming risk	Can be reduced by concentrating tail risk	Cannot be gamed in this way

Table 1.1: Value at Risk and Expected Shortfall: a comparison

1.3 Volatility Dynamics in Equity Markets

Both VaR and ES depend critically on the assumed return distribution, and the most important parameter of that distribution is the conditional variance σ_t^2. The empirical literature on equity returns has established three regularities that any serious risk model must confront.

Fat tails. Mandelbrot (1963) observed that speculative price distributions exhibit leptokurtosis: extreme returns occur far more often than the Normal distribution implies. This property is present in virtually every major equity index studied in the literature. In the S&P 500 weekly return data used in this thesis, the excess kurtosis is 7.94 and standard tests reject normality decisively. The practical consequence is that Normal-distribution based risk models systematically understate the probability and severity of extreme losses. At the 99th percentile, t-Student VaR with moderate degrees of freedom can be 20–30% larger than their Normal equivalents. The t-Student distribution, with its shape governed by an estimated degrees-of-freedom parameter ν, is the natural parametric alternative to the Normal for high-confidence level risk measurement.

Volatility clustering.-A common observation in financial markets is that large returns tend to be followed by large returns - not necessarily of the same sign, but of similar magnitude. The autocorrelation function of raw returns is near zero at all lags, confirming the absence of return predictability, but the ACF of squared returns shows strong, slowly decaying positive dependence at all lags up to one year. This predictable structure in variance – illustrated in next chapters for the S&P 500 sample - is precisely what EWMA and GARCH models exploit. A model that treats variance as constant over time will overstate risk during calm periods and understate it during turbulent ones, with systematic errors exactly when accurate estimation matters most.

The leverage effect. Black (1976) documented that negative return shocks increase future volatility more than positive shocks of equal magnitude. This asymmetry, which is labelled as the leverage effect because a declining equity price raises the firm's debt-to-equity ratio, implies that the conditional distribution following a market decline has a heavier left tail than following a rally. The standard GARCH (1,1) responds symmetrically to shocks of either sign, but the EGARCH specification of Nelson (1991) captures this asymmetry through a signed shock term. Whether this asymmetry materially improves VaR and ES calibration at weekly frequency remains an open empirical question and is left for future research.

II. METHODOLOGY

This chapter describes the models and procedures used to estimate and backtest VaR and ES. Section 2.1 presents the EWMA model. Section 2.2 develops GARCH (1,1). Section 2.3 derives the VaR and ES formulas under both distributional assumptions. Section 2.4 describes the four backtesting procedures. All models are applied to weekly S&P 500 log-returns, which are downloaded from stooq.pl, over the period January 2007 - March 2026. A full description of the sample is provided in Section 3.1. All models are estimated using a rolling window of 250 weeks, re-estimated each period as the window advances one step through the out-of-sample period.

2.1 The EWMA Model

The Exponentially Weighted Moving Average model was introduced in the RiskMetrics technical document of J.P. Morgan/Reuters (1996) as a computationally tractable approach to time-varying volatility estimation. The core idea is simple: rather than giving equal weight to all past observations as a sample variance does, EWMA assigns geometrically declining weights to observations as they recede into the past, so that recent observations influence the current variance estimate more than distant ones.

The EWMA variance estimate at time t is defined by the recursion:

$$\sigma_t^2 = \lambda\sigma_{t-1}^2 + (1 - \lambda)r_{t-1}^2$$

where r_{t-1} is the return in the previous period and $\lambda \in (0, 1)$ is the **decay parameter**. Substituting the recursion repeatedly, the estimate can be written as an infinite weighted sum of past squared returns:

$$\sigma_t^2 = (1 - \lambda)\sum_{j=0}^{\infty} \lambda^j r_{t-1-j}^2$$

The weight assigned to an observation *j* period ago is $(1 - \lambda)\lambda^j$, which declines geometrically at rate λ. The effective memory of the model — defined here as the number of lags for which the assigned weight exceeds 1% of the weight on the most recent observation — is approximately $\frac{1}{1-\lambda}$. For the RiskMetrics standard of λ = 0.94 applied to weekly data, this corresponds to roughly 17 weeks: a shock from more than four months ago retains less than 10% of the weight assigned to the most recent observation.

Three structural properties of EWMA are important for interpreting the backtesting results in Chapter 3. First, EWMA requires **no parameter estimation:** given the fixed λ, the variance is updated by a single recursive operation each period. This makes the model robust to estimation error and operationally straightforward. Second, the EWMA process is **integrated**: the coefficients on σ_{t-1}^2 and r_{t-1}^2 sum to one, so the process has no unconditional variance and shocks persist indefinitely with no pull toward a long-run variance level. A volatility spike today will decay gradually but never be 'corrected' back to a fixed baseline. Third, EWMA responds **symmetrically** to positive and negative shocks: a sharp rally raises the variance estimate by exactly as much as a crash of the same magnitude.

2.2 The GARCH (1,1) Model

The Generalized Autoregressive Conditional Heteroskedasticity model was introduced by Bollerslev (1986) as a parsimonious extension of the ARCH framework of Engle (1982). In contrast to EWMA, GARCH estimates its parameters from the data and incorporates reversion to equilibrium variance — the tendency for conditional variance to return toward its long-run unconditional level after a shock. These differences are not incidental: they reflect a fundamentally different view of how volatility should be modelled and have direct consequences for backtesting performance across different market regimes.

The GARCH (1,1) conditional variance specification is:

$$\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}$$

where $\omega > 0, \alpha \geq 0, \beta \geq 0$, and the stationarity condition $\alpha + \beta < 1$ must hold. The parameter ω is the long-run variance floor; α governs how strongly the current variance responds to the most recent squared return (shock sensitivity); β determines how persistent variance is from period to period. The sum $\alpha + \beta$ is the overall **persistence** of the variance process: values close to one, typical for equity returns, imply that volatility shocks are long-lived. When $\alpha + \beta < 1$, the GARCH (1,1) process is covariance-stationary and has a well-defined unconditional variance:

$$\sigma^{-2} = \frac{\omega}{(1-\alpha-\beta)}$$

This is the level to which conditional variance eventually reverts after a shock. The speed of reversion is governed by $\alpha + \beta$: the **half-life** of a volatility shock - the time after which half of its impact has dissipated - is $ln(0.5) \,/\, ln(\alpha + \beta)$. For typical equity return series, estimated persistence values of 0.90 to 0.97 imply half-lives ranging from roughly 7 to 23 weeks, meaning that a volatility spike from a market shock will still be partly visible in the variance estimate for several months.

GARCH model parameters are estimated by **Maximum Likelihood Estimation (MLE)**. Under the assumption of conditionally Normal innovations — that is, assuming the standardized return $\varepsilon_t = \frac{r_t}{\sigma_t}$ follows a standard Normal distribution — the log-likelihood function is:

$$\ell(\omega, \alpha, \beta) = -\frac{T}{2}\ln(2\pi) - \frac{1}{2}\sum_{t=1}^{T}[\ln(\sigma_t^2) + \frac{r_t^2}{\sigma_t^2}]$$

When t-Student innovations are assumed, the density is modified accordingly and the degrees-of-freedom parameter ν is estimated jointly with (ω, α, β). The estimated ν characterises the fat-tail behaviour of the standardised residuals, and a smaller ν indicates heavier tails and a greater departure from Normality.

The **key structural difference between GARCH and EWMA** is about variance in equilibrium σ^{-2} . Once a volatility spike occurs, GARCH pulls the conditional variance back toward σ^{-2} at a speed determined by $\alpha + \beta$, while EWMA simply lets it decay at the fixed rate λ. This difference proves consequential in the stress-test analysis of Section 3.4: during the 2022 interest rate shock - a prolonged, moderate-severity stress episode - GARCH's higher α coefficient causes it to over-react to each individual weekly loss, while EWMA's smoother decay tracks the sustained stress more effectively.

2.3 VaR and ES Estimation Formulas

Once the conditional variance σ_t^{-2} is estimated, VaR and ES are obtained by combining it with a distributional assumption for the standardized innovation $\varepsilon_t = \frac{r_t}{\sigma_t}$ The conditional mean $\hat{\mu}$ is set to the rolling sample mean, which at weekly frequency contributes negligibly to the risk estimates relative to the variance term.

Normal distribution. Under the assumption that the standardized return $\varepsilon_t = \frac{r_t - \hat{\mu}}{\sigma_t}$ follows a standard Normal distribution, the one-period-ahead VaR and ES at confidence level α are:

$$VaR_\alpha = -\hat{\mu} + \sigma_t \Phi^{-1}(\alpha)$$

$$ES_\alpha = -\hat{\mu} + \sigma_t \, \varphi \frac{\left(\Phi^{-1}(\alpha)\right)}{(1-\alpha)}$$

where $\Phi^{-1}(\cdot)$ is the quantile function of the standard Normal distribution and $\varphi(\cdot)$ is its density function. The quantity $\varphi \frac{\left(\Phi^{-1}(\alpha)\right)}{(1-\alpha)}$ is the *ES multiplier*, i.e. a factor by which σ_t is scaled to produce the ES estimate. Under the Normal distribution, this multiplier is fixed: it equals 2.063 at α = 95% and 2.665 at α = 99%, regardless of the level of volatility. VaR and ES are therefore always in a fixed proportional relationship under the Normal assumption, which is one reason the Normal distribution tends to understate the gap between the two measures when returns are fat-tailed.

Student's t-distribution. Under the assumption that ε_t follows a standardised t-distribution with ν degrees of freedom, standardized so that the variance equals one, the VaR and ES are:

$$VaR_\alpha = -\hat{\mu} + \sigma_t \sqrt{\frac{\nu - 2}{\nu}} \, t_{\alpha,\nu}^{-1}$$

$$ES_\alpha = -\hat{\mu} + \sigma_t \sqrt{\frac{\nu - 2}{\nu}} \, \frac{f_\nu\left(t_{\alpha,\nu}^{-1}\right)\left(\nu + t_{\alpha,\nu}^{-1\,2}\right)}{(\nu - 1)(1-\alpha)}$$

The factor $\sqrt{\frac{\nu-2}{\nu}}$ ensures unit variance of standardized residuals, consistent with the GARCH parameterization. Unlike the Normal case, the t-Student ES multiplier depends on ν and grows substantially as ν decreases (tails become heavier) and as the confidence level increases. At $\nu = 9.17$ and $\alpha = 99\%$, the t-Student ES multiplier (3.21) is 20% larger than the Normal multiplier (2.67). This difference directly determines whether a model passes or fails the ES backtesting criterion in Chapter 3.

Confidence level α	Normal quantile $\Phi^{-1}(\alpha)$	t-quantile (ν = 9)	Ratio t / Normal	Normal ES multiplier	t-Student ES multiplier
95%	1.645	1.406	0.86	2.063	2.257
99%	2.326	2.647	1.14	2.665	3.969
99.5%	2.576	3.363	1.31	2.892	4.986

Table 2.1: VaR quantile comparison — Normal vs. t-Student at selected confidence levels

The pattern in Table 2.1 reveals an asymmetric relationship between the Normal and t-Student distributions across confidence levels. At 95%, the t-Student produces a *lower* VaR multiplier (1.406 vs 1.645), suggesting the Normal distribution overstates risk at moderate confidence levels. However, this reverses sharply at higher quantiles: at 99%, the t-Student VaR exceeds the Normal by 14% (2.647 vs 2.326), and at 99.5% the gap widens to 31% (3.363 vs 2.576). The discrepancy in ES multipliers is even more pronounced — at 99% the t-Student ES multiplier (3.969) is 49% larger than the Normal (2.665), and at 99.5% it is 72% larger (4.986 vs 2.892). This pattern reflects the fat-tailed nature of the t-Student distribution: it assigns substantially more probability mass to extreme losses than the Normal. For risk management purposes, the direction of the error matters — at the high confidence levels used in regulatory VaR (99%) and stressed ES (99.5%), relying on the Normal distribution **understates** risk capital requirements significantly. This is the quantitative case for preferring the t-Student assumption at high confidence levels.

2.4 Backtesting Framework

Backtesting evaluates whether a model's risk forecasts are statistically consistent with subsequent realized returns. A **violation** occurs whenever the realized return falls below the negative of the VaR forecast: $\eta_t = 1\{r_t < -VaR_{\alpha,t}\}$. The estimation procedure uses a **rolling window of 250 weeks**: at each point in the out-of-sample period, the model is re-estimated on the 250 most recent observations and $VaR_{\alpha,t}$ and $ES_{\alpha,t}$are forecast for the next period. The window then advances one step. Let be total number of violations, n_0 the number of non-violations, and $T = n_0 + n_1$ the total out-of-sample observations.

Test 1. Kupiec (1995) Unconditional Coverage.

The Kupiec test, also known as the Proportion of Failures (POF) test, evaluates whether the observed violation frequency is statistically consistent with the confidence level stated by the model. The theoretical violation probability p is equal to 0.05 for the 95% confidence level and 0.01 for the 99% level. The empirical violation rate is $\pi = \frac{n1}{(n_0 + n_1)}$.

The null hypothesis of the test is:

$$H_0 : P(\eta_t = 1) = p$$

The test is based on the likelihood ratio comparing the likelihood of the data under the null (probability p) to the likelihood under the unrestricted estimate (probability π):

$$LR_{UC} = \frac{p^{n1}(1-p)^{n_0}}{\pi^{n_1}\,(1-\pi)^{n_0}}$$

Under the null, test statistics has asymptotic Chi-square distribution: $-2ln\ (LR_{UC}) \sim \chi^2(1)$. Rejection of the null indicates the observed violation rate is inconsistent with the confidence level.

Test 2. Christoffersen (1998) Independence Test (Ch2)

A model can pass the Kupiec test while clustering all violations in one episode, which indicates a failure to track changing volatility. The Ch2 test isolates this by testing whether consecutive violations are independent. Under the null hypothesis, we assume that the VaR violation at time t is independent of the violation at time $t-1$:

$$H_0 : P(\eta_t = 1 | \eta_{\{t-1\}} = 0) = P(\eta_t = 1 | \eta_{\{t-1\}} = 1) = p$$

Denote by n_{ij} the number of observations for which

$$n_{\{ij\}} = \#\ (\eta_{\{t-1\}} = i\ \ and\ \ \eta_t = j)$$

Denote by π_0 and π_1 the ratios of violations after non-violation and after violation respectively and by $\pi_0 = \frac{n_{01}}{n_{00}+n_{01}}\ and\ \pi_1 = \frac{n_{11}}{n_{10}+n_{11}}$ the unconditional violation rate. The likelihood ratio statistic is

$$LR_{Ch2} = \frac{p^{n_{01}+n_{11}}(1-p)^{n_{00}+n_{10}}}{\pi^{n_{01}}(1-\pi)^{n_{00}}\pi^{n_{11}}(1-\pi)^{n_{10}}}$$

Under the null, test statistics $-2\ln(LR_{Ch2})$ is asymptotically $\chi^2(1)$ distributed. Rejection of this test indicates that the probability of a violation today depends on whether yesterday a violation was also — a sign of clustering that a well-specified dynamic model should not exhibit.

Test 3. Christoffersen and Pelletier (2004) Duration.

Rather than examining consecutive violations, the Ch-P test looks at the time between violations. Under the null of correct model specification, the durations $d_i = \tau_i - \tau_{i-1}$ between consecutive violations should follow an exponential distribution, exhibiting no memory of when the last violation occurred. Clustering implies a Weibull distribution with shape parameter $\alpha \neq 1$. Under the null of Ch-P test, $\alpha = 1$ (Weibull reduces to Exponential). The test is more powerful than Ch2 against violations that cluster over multiple consecutive periods, as is common during sustained crisis episodes. Under the null, test statistics is $\chi^2(1)$ distributed.

Test 4. McNeil and Frey (2000) for Expected Shortfall.

The three tests above evaluate VaR only. The McNeil and Frey test extends the backtesting framework to ES by examining whether the magnitude of losses in violation periods is consistent with the model's ES forecast. For each period τ in which a violation occurs, the exceedance residual is defined as:

$$e_\tau = \frac{r_\tau - ES_{\alpha,\tau}}{\sigma_\tau}$$

Under the null hypothesis of correct ES specification, these residuals should have mean zero - the actual tail loss on average equals the predicted ES. The test statistic is the t-ratio of the mean exceedance residual, with p-values obtained by bootstrap resampling. Rejection means actual losses in violation periods are systematically larger than the model predicts. This test is the direct operationalization of the regulatory question that motivated the shift from VaR to ES under Basel III.

A model is assessed to pass backtesting if all four tests simultaneously fail to reject at the 5% significance level.

III. EMPIRICAL STUDY

This chapter presents the full empirical analysis. Section 3.1 describes the data and documents the key statistical properties of the S&P 500 return series. Section 3.2 reports model estimation results. Section 3.3 presents backtesting results for all 14 model–distribution combinations at both confidence levels. Section 3.4 disaggregates performance across four distinct market regimes. Section 3.5 synthesizes the findings and situates them in the context of the existing literature.

3.1 Data Description and Descriptive Statistics

The empirical analysis is based on weekly log-returns of the S&P 500 Composite Index, sourced from stooq.pl using the ticker ^*spx* with weekly frequency. The sample covers the period from January 2007 to March 2026, yielding approximately 1,000 weekly observations. Log-returns are computed as $r_t = \ln(P_t/P_{t-1})$, where P_t is the weekly closing level. The sample start date of January 2007 is chosen to include the Global Financial Crisis of 2008–2009, which represents the most severe equity market stress episode of the past two decades. The sample ends in March 2026, incorporating the COVID-19 market crash of 2020 and the interest rate shock of 2022 - three qualitatively distinct stress episodes that test models under very different conditions.

As illustrated in Figure 3.1, over the sample period, the index rose from approximately 1,400 points at the start of 2007 to around 7,000 points by early 2026, reflecting the long-run upward trend in US equities. However, this overall growth conceals three major drawdown episodes: a decline of roughly 57% during the GFC of 2008–2009, a collapse of approximately 34% in five weeks during the COVID-19 crash of February–March 2020, and a more gradual bear market of roughly 25% over the course of 2022 driven by aggressive interest rate increases. These three episodes serve as the stress-test benchmarks throughout the analysis.

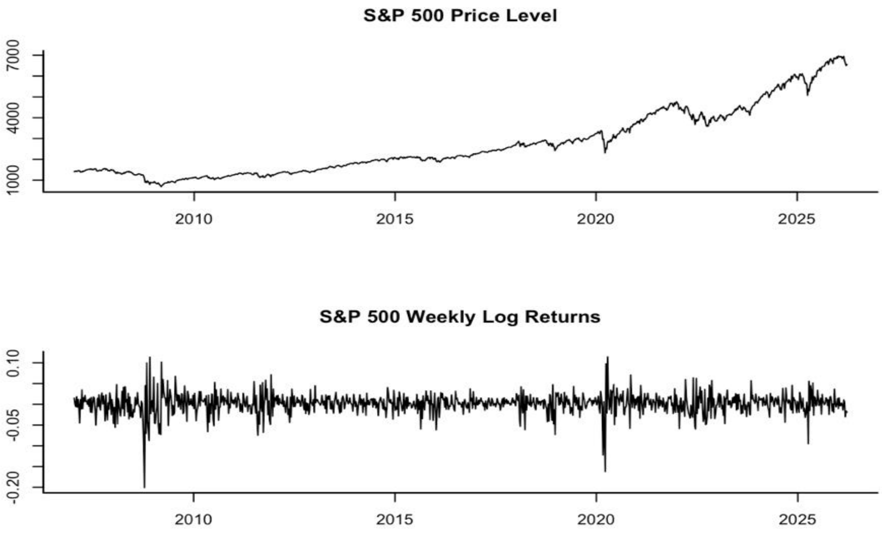

Figure 3.1: S&P 500 price level (above) and weekly log-returns (below). The three major stress episodes are evident: GFC October 2008 (−20% single week), COVID-19 March 2020 (−18% single week), and the 2022 rate-hike bear market (sustained losses over ~40 weeks). Return volatility clustering is visible throughout.

Statistic	Value	Interpretation
Mean	0.001526	~0.15% per week; 7.96% annualised
Standard deviation	0.025254	2.5% per week; 18.24% annualised
Minimum	−0.2004	Worst week: −20.0% (October 2008, GFC)
Maximum	+0.1196	Best week: +11.96% (November 2008, GFC relief)
Skewness	−0.9386	Negative: extreme losses exceed extreme gains
Kurtosis	10.9429	Excess kurtosis 7.94 — very fat tails
t-Student df (MLE)	9.166	Moderate fat tails; fits data substantially better than Normal
Normality tests	$p < 2.2\times10^{-16}$	All three tests reject normality decisively

Table 3.1: Descriptive statistics of S&P 500 weekly log-returns, January 2007 – March 2026

Table 3.1 presents the descriptive statistics of the return series. Four features motivate the modelling choices. The negative skewness of −0.94 and excess kurtosis of 7.94 confirm the two stylized facts identified in Chapter 1. Extreme losses dominate extreme gains in magnitude and returns far beyond three standard deviations occur approximately eight times more often than the Normal distribution predicts. The t-Student distribution with $\nu = 9.17$, estimated by maximum likelihood, fits the data substantially better: the AIC improves by 102.6 points, the KS test no longer rejects the null ($p = 0.412$), and the QQ plot against the t-Student distribution eliminates the pronounced S-shape that characterises the Normal QQ plot.

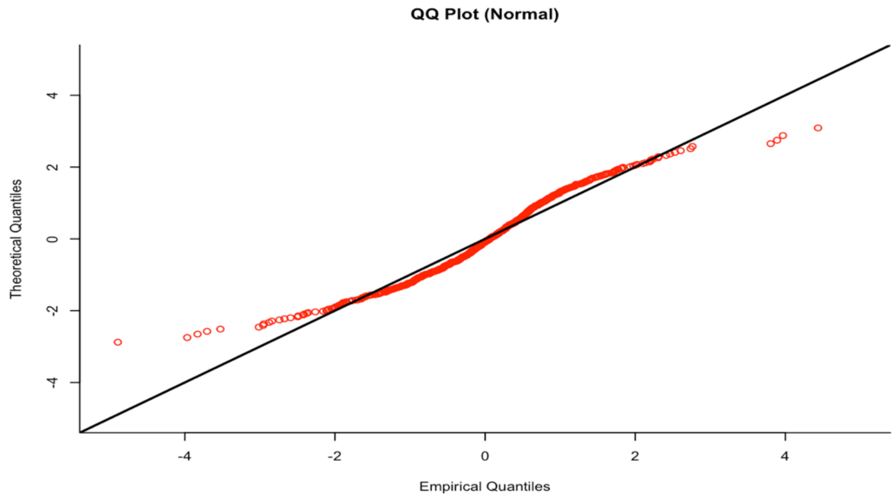

Figure 3.2 (above panel): QQ plot of S&P 500 weekly log-returns against the Normal distribution. The S-shaped deviation indicates excess kurtosis: the left tail extends to approximately −4.5 standard deviations while the Normal predicts only −3.0 at the same probability level.

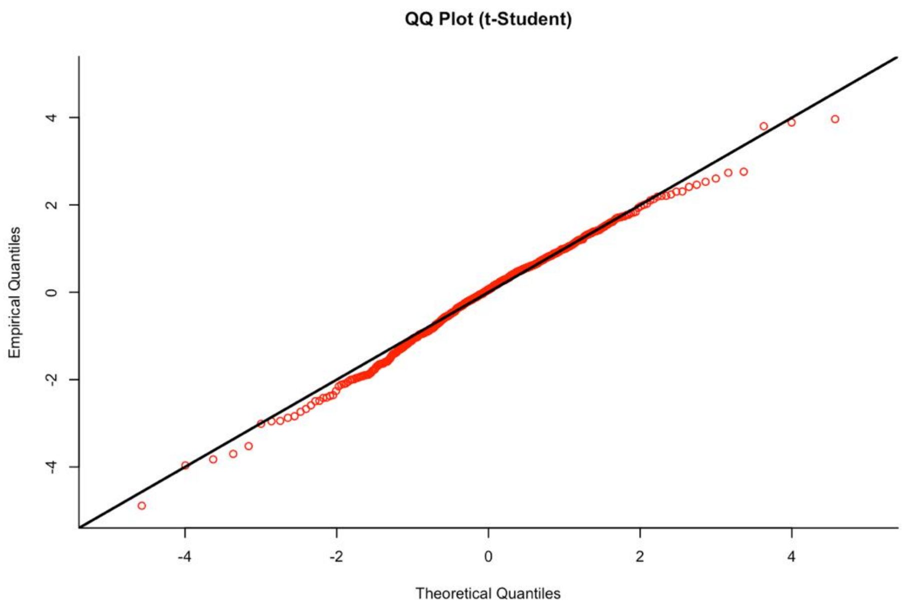

Figure 3.2 (below panel): QQ plot against the t-Student distribution (ν = 9.17). The bulk of the distribution aligns closely with the reference line, confirming the t-Student as a substantially better fit than the Normal. Minor deviations remain in the extreme tails — the left tail falls slightly below the line, indicating that the most severe observed losses are marginally heavier than the fitted distribution implies — but these are consistent with the occasional outliers expected in financial return data.

The ACF plot (Figure 3.3) provides direct evidence for volatility clustering. The ACF of raw returns is near zero at all lags - consistent with return unpredictability - while the ACF of squared returns is significantly positive and decays slowly over 20+ weeks. This pattern is typical for ARCH structure and is the primary empirical motivation for conditional volatility models.

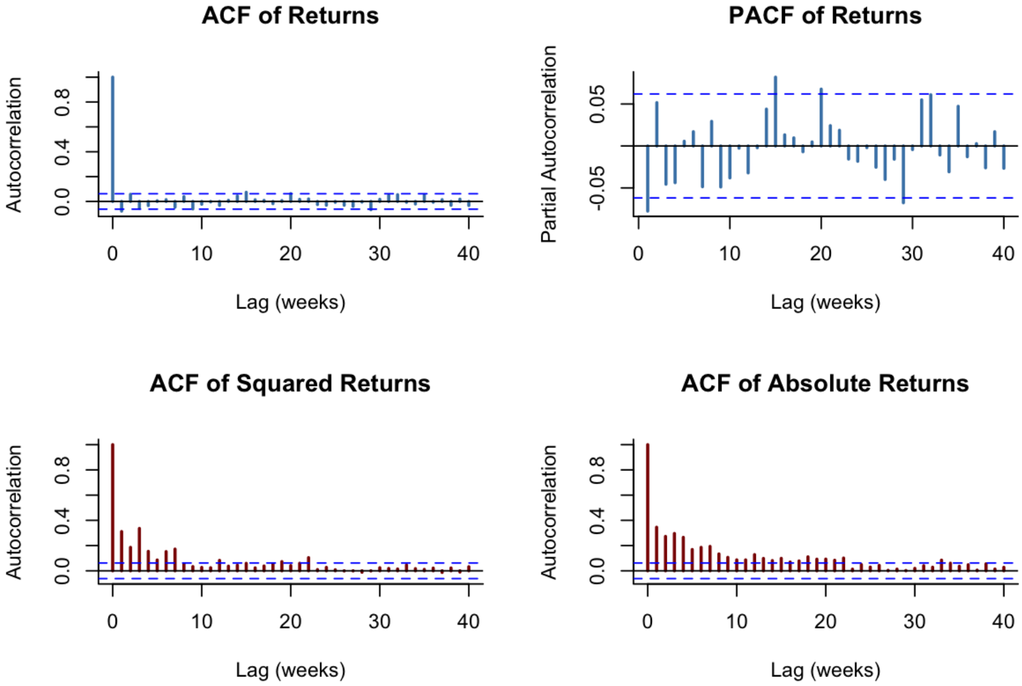

Figure 3.3: Autocorrelation analysis. Near-zero ACF of raw returns confirms the absence of linear predictability. The significant, slowly decaying ACF of squared returns confirms time-varying conditional variance. Dashed lines are 95% confidence bands under the null of white noise.

3.2 Model Estimation Results

Before backtesting, each model is estimated on the full sample to characterise the volatility dynamics of the S&P 500 return series. This section reports the unconditional distribution fit and the GARCH (1,1) parameter estimates.

3.2.1 Unconditional VaR and Expected Shortfall

The unconditional VaR and ES estimates in Table 3.2 serve as the static baseline against which the dynamic models are compared. Three distributional assumptions are considered: Normal, t-Student with $\nu = 9.17$, and Historical Simulation. The estimates are computed at the 95% and 99% confidence levels using the full-sample parameter estimates.

Model	VaR 95%	VaR 99%	ES 95%	ES 99%
Normal distribution	−4.31%	−6.03%	−5.06%	−6.58%
t-Student (ν = 9.17)	−3.71%	−6.84%	−5.55%	−9.88%
Historical Simulation	−3.98%	−7.12%	−6.21%	−10.46%

Table 3.2: Unconditional VaR and ES estimates

At the 95% confidence level, all three models agree closely (VaR range from −3.93% to −4.00%), confirming that distributional choice has minimal impact at moderate quantiles. The divergence grows substantially at 99%: the t-Student VaR of −6.13% is 40 basis points larger than the Normal VaR, and the Historical ES of −10.46% is 59% larger than Normal ES. This gap directly reflects the GFC and COVID crashes physically present in the empirical tail - observations that the thin-tailed Normal distribution cannot reproduce. The relationship between Normal and t-Student estimates here is consistent with the multiplier comparison in Table 2.1: at 99%, the t-Student quantile exceeds the Normal by approximately 21%, broadly matching the 40-bps gap.

3.2.2 GARCH (1,1) Parameter Estimates

For the GARCH (1,1) model, the parameters are estimated by maximum likelihood over the full sample. Table 3.3 reports the estimates under both distributional assumptions.

Parameter	GARCH Normal	GARCH t-Student	Interpretation
ω (constant)	0.000049	0.000031	Long-run variance contribution
α (ARCH effect)	0.3196	0.2766	Sensitivity to recent shocks
β (GARCH effect)	0.6301	0.6772	Persistence of past variance
α + β (persistence)	0.9497	0.9538	Overall variance persistence
ν (t-Student df)	—	6.6224	Fat-tail parameter (MLE)

Table 3.3: GARCH (1,1) parameter estimates

The persistence $\alpha + \beta = 0.9497$ (Normal) and 0.9538 (t-Student) imply half-lives of 13.4 and 14.7 weeks respectively. A volatility spike will therefore still be partially visible in the conditional variance estimate after more than three months. This slow mean reversion is consistent with the slowly decaying ACF of squared returns documented in Figure 3.3. Note that the estimated degrees-of-freedom parameter under the GARCH t-Student specification is $\nu = 6.62$**,** which differs from the full-sample unconditional estimate of $\nu = 9.17$ in Section 3.1. This is expected: the GARCH model conditions on time-varying volatility, so the standardized residuals $\varepsilon_t = r_t/\sigma_t$ have thinner tails than the raw returns, resulting in a higher estimated ν. The lower ν = 6.62 from the GARCH MLE reflects the tail behavior of the standardized innovations after volatility clustering has been removed.

The comparison with EWMA is instructive. EWMA's effective ARCH weight is $1 - \lambda = 0.06$, while GARCH Normal's $\alpha = 0.3196$ is more than five times larger and GARCH t-Student's $\alpha = 0.2766$ is more than four times larger. GARCH therefore reacts far more strongly to each individual weekly return shock. This reactivity is visible in Figure 3.4: the GARCH conditional volatility series is noticeably more jagged than EWMA's, with sharper spikes following individual extreme weeks. During the 2022 rate shock — a sustained sequence of moderate weekly losses rather than a single extreme event — this higher reactivity becomes a liability, as GARCH over-responds to each individual loss and generates more violations than EWMA. This distinction motivates the regime-specific analysis in Section 3.4.

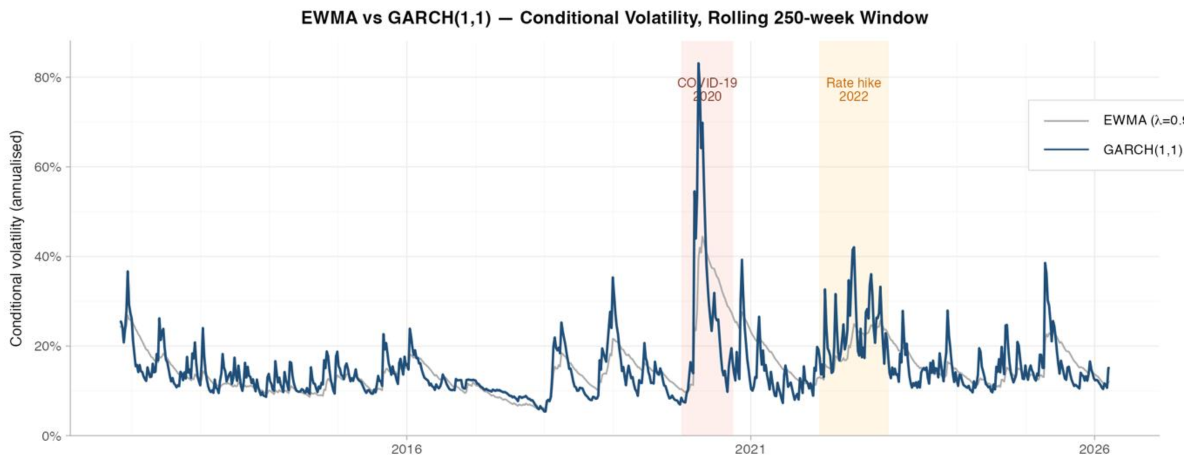

Figure 3.4: Annualized conditional standard deviation from EWMA (grey, λ=0.94) and GARCH (1,1) t-Student (blue), rolling 250-week estimation window. Both series spike during the COVID-19 crash (March 2020) and show elevated readings during the 2022 rate cycle. The GARCH estimate is visibly more jagged, reflecting its higher ARCH coefficient (α=0.32 vs. EWMA's implied 0.06).

3.3 Backtesting Results

All backtesting is conducted on approximately 730 out-of-sample weeks from 2012 to March 2026, using a 250-week rolling estimation window. The out-of-sample period therefore includes the calm bull market of 2013–2019, the COVID-19 crash of 2020, the rate-hike period of 2022, and the subsequent stabilization phase — a range of market environments that provides a demanding test of model adequacy. Results are reported for all 14 model–

distribution combinations. **PASS requires all four tests to simultaneously not reject at the 5% level.**

3.3.1 Unconditional Models

Model	Level	Exp.	Act.	Kupiec	Ch2	Ch-P	ES test	Overall
Normal	95%	38	24	0.003 ✗	0.001 ✗	0.143 ✗	0.000 ✗	FAIL
t-Student	95%	38	24	0.093 ✗	0.003 ✗	0.032 ✗	0.000 ✗	FAIL
Historical	95%	38	24	0.015 ✗	0.007 ✗	0.041 ✗	0.455 ✓	FAIL
Normal	99%	8	7	0.325 ✓	0.044 ✓	0.000 ✓	0.000 ✗	PARTIAL
t-Student	99%	8	5	0.325 ✓	0.044 ✓	0.000 ✓	0.000 ✗	PARTIAL
Historical	99%	8	5	0.325 ✓	0.044 ✗	0.000 ✗	0.389 ✓	PARTIAL

Table 3.4: Backtesting results - unconditional models. Kupiec, Ch2, Ch-P, and ES columns report p-values of the respective test statistics. ✓ = fail to reject at 5% (p > 0.05); ✗ = reject at 5% (p ≤ 0.05).

Table 3.4 reports backtesting results for the unconditional models. The Normal and t-Student specifications produce 21 and 28 violations respectively at the 95% level against an expected 38, while Historical Simulation produces 24, failing the Kupiec, Ch2, and Ch-P tests simultaneously. The problem is systematic and well understood in the literature: unconditional models estimated on a sample including the volatile GFC and COVID episodes produce variance estimates — and therefore VaR thresholds — calibrated to crisis-era conditions. During the predominantly calm out-of-sample period of 2013–2019, these thresholds are far too conservative, generating too few violations. This finding is consistent with the results of McNeil et al. (2015), who document that static model systematically over-estimate risk during low-volatility regimes.

At 99%, the pattern improves partially. Normal and t-Student pass all three VaR tests: the strict threshold means that even a model calibrated to crisis volatility generates approximately the right number of deep-tail violations over a long sample. However, both fail the ES test: the Normal distribution ES (99%) fails the ES test decisively ($p < 0.001$), as its thin tail systematically understates average losses in violation periods. The t-Student specification also fails the ES test ($p < 0.001$), reflecting that even the fat-tailed distribution cannot adequately capture the deep tail losses when the VaR threshold is too conservative. Historical Simulation passes Kupiec and the ES test at 99% but fails Ch2 and Ch-P, reflecting clustering of its violations around the COVID episode. **No unconditional model achieves a full PASS at any confidence level.**

3.3.2 EWMA Model

Model	Level	Exp.	Act.	Kupiec	Ch2	Ch-P	ES test	Overall
EWMA Normal	95%	38	32	0.337 ✓	0.278 ✓	0.096 ✓	0.002 ✗	PARTIAL
EWMA Normal	99%	8	15	0.016 ✗	0.005 ✗	0.047 ✗	0.003 ✗	FAIL
EWMA t-Student	95%	38	32	0.337 ✓	0.278 ✓	0.096 ✓	0.006 ✗	PARTIAL
EWMA t-Student	99%	8	15	0.016 ✗	0.005 ✗	0.047 ✗	0.049 ✗	FAIL

Table 3.5: Backtesting results — EWMA model

Table 3.5 reports results for EWMA. At 95%, both EWMA specifications pass all three VaR tests: the actual violation counts of 32 is close to the expected 38, and the Ch-P test (p = 0.096) confirms that violations are not significantly clustered in time. This improvement over unconditional models is entirely attributable to the rolling-window estimation: EWMA tracks the declining volatility of the 2013–2019 calm period and lowers its VaR threshold, accordingly, generating violations at the appropriate frequency. The ES test, however, fails decisively for EWMA Normal (p = 0.002): when violations do occur, the losses substantially exceed the model's ES estimate. This confirms that the failure is distributional, not structural - the Normal tail is too thin even when the volatility level is correctly estimated. Switching to t-Student improves the ES p-value to 0.006, a meaningful improvement, but still falls below the 5% threshold.

At 99%, EWMA fails comprehensively under both distributions. The violation counts of 15 is nearly double the expected 8, and all four tests reject. The most revealing comparison is between the two distributional assumptions on the ES test: the p-value improves from 0.003 (Normal) to 0.049 (t-Student) - a 17-fold improvement that **nearly crosses the 5% threshold**. This confirms the second research hypothesis in the negative direction: the distributional assumption is the binding constraint for ES accuracy at 99%, even when the volatility model is correctly specified. The fixed decay structure of EWMA (no mean reversion, λ = 0.94) is insufficient to generate the extreme variance estimates needed for adequate 99% VaR coverage during sudden shock episodes.

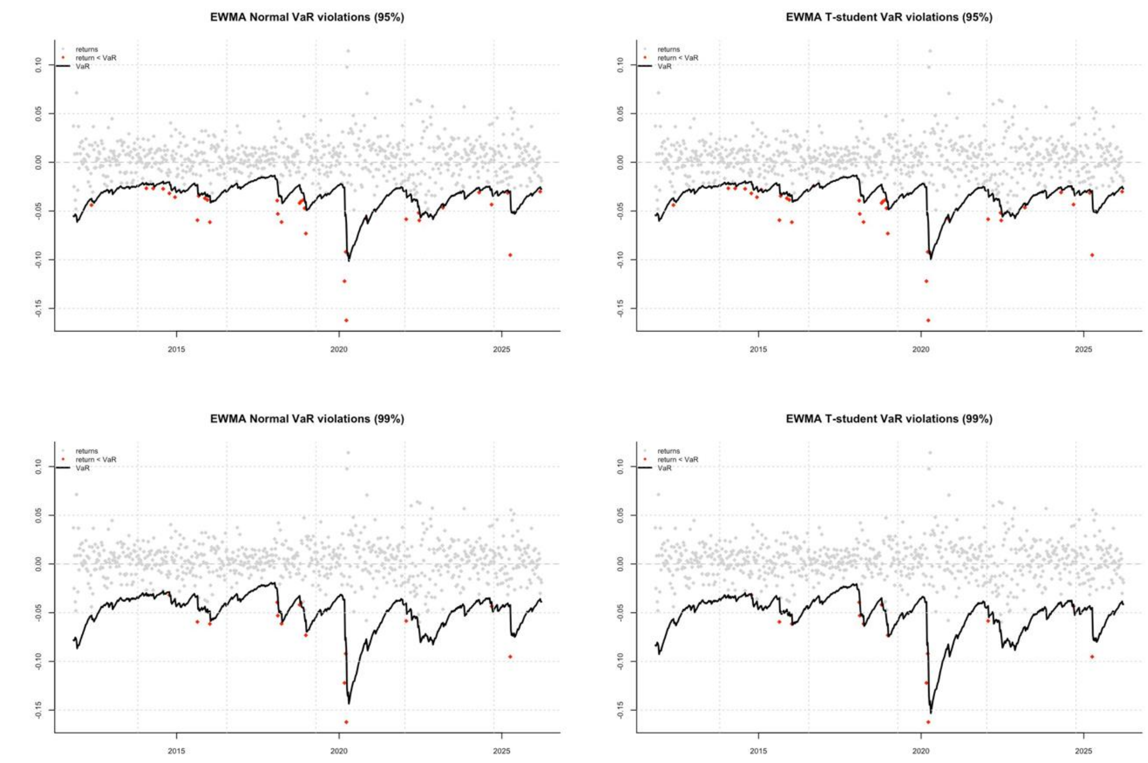

Figure 3.5: S&P 500 weekly returns against EWMA Normal VaR thresholds at 95% and 99%. The dynamic VaR line (black) rises sharply during the COVID-19 crash, demonstrating EWMA's responsiveness to sudden volatility increases. Red dots mark violations. Note the excessive violation cluster at 99% during 2020

3.3.3 GARCH (1,1) Model

Model	Level	Exp.	Act.	Kupiec	Ch2	Ch-P	ES test	Overall
GARCH Normal	95%	38	50	0.048 ✗	0.041 ✗	0.177 ✓	0.170 ✓	PARTIAL
GARCH Normal	99%	8	15	0.016 ✗	0.032 ✗	0.631 ✓	0.047 ✗	PARTIAL
GARCH t-Student	95%	38	53	0.015 ✗	0.030 ✗	0.149 ✓	0.624 ✓	PARTIAL
GARCH t-Student	99%	8	13	0.069 ✓	0.089 ✓	0.948 ✓	0.154 ✓	PASS ⭑

Table 3.6: Backtesting results — GARCH (1,1) model

At 95%, GARCH generates too many violations - 50–53 against an expected 38 - and fails the Kupiec and Ch2 tests. This is the opposite failure to EWMA at 95% and arises from GARCH's high ARCH coefficient: each individual weekly shock immediately raises the conditional variance, lowering the VaR threshold and increasing the chance that a subsequent week’s return will be classified as a violation. However, GARCH passes both the Ch-P and ES tests at 95%. The Ch-P result (p = 0.177 and p = 0.149) confirms that the excess violations are not temporally clustered – GARCH’s equilibrium variance-reversion structure prevents the persistence of elevated variance that would cause clusters. The ES test results (p = 0.170 and p = 0.624) confirm that the magnitude of losses in violation periods is consistent with the model's ES forecasts. The GARCH t-Student ES p-value of 0.624 is the strongest ES result in the entire analysis - strong evidence that the t-Student tail correctly calibrates expected tail losses at the 95% level.

At 99%, GARCH t-Student is the only model in the entire analysis to achieve a full PASS. The actual violation counts of 13 against an expected 8 produces a Kupiec p-value of 0.069 — just above the 5% threshold. The Ch2 test passes with p = 0.089, confirming violation independence. Most strikingly, the Ch-P test passes with p = 0.948: the durations between violations are nearly perfectly consistent with the memoryless exponential distribution, indicating that violations are scattered randomly through time rather than clustering in any episode. The ES test passes with p = 0.154, confirming adequate tail loss calibration. The t-Student innovation distribution is essential to this result: GARCH Normal fails the ES test at 99% (p = 0.047) with otherwise almost identical VaR characteristics. This directly confirms the second and third research hypotheses: the correct distributional tail assumption (t-Student) and the correct volatility dynamics (GARCH reversion to equilibrium variance) are jointly necessary conditions for adequate 99% performance.

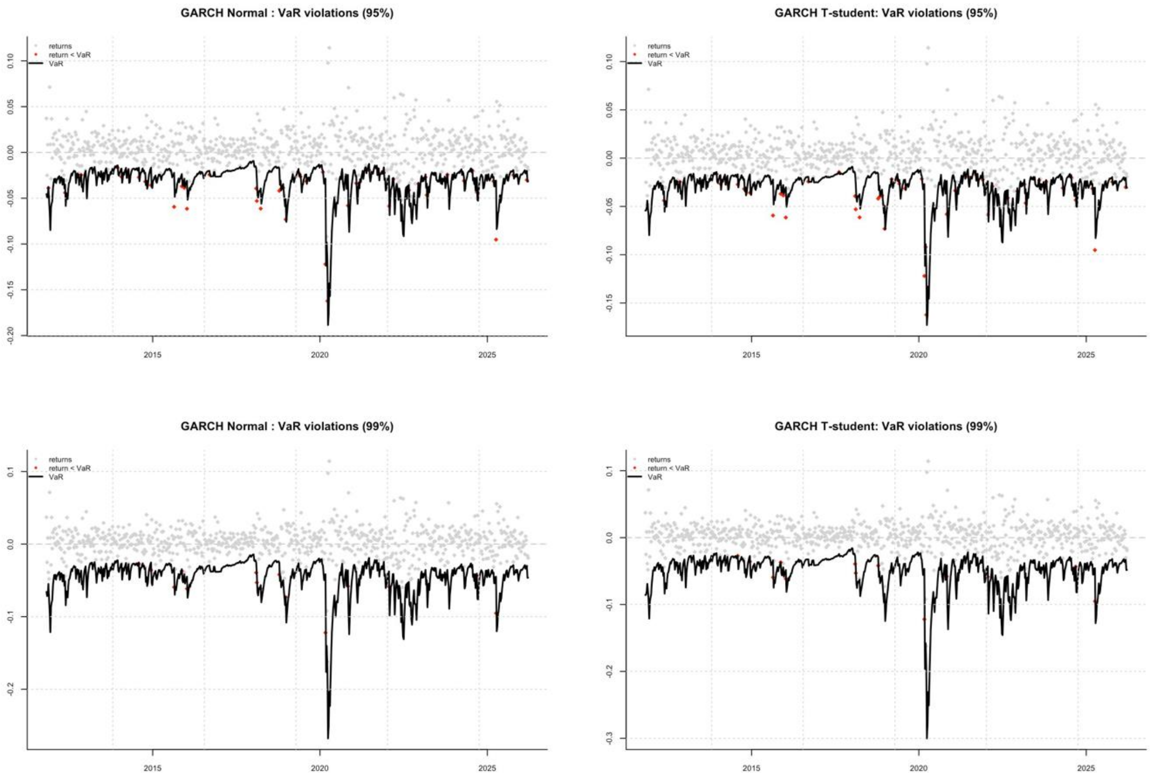

Figure 3.6: S&P 500 weekly returns against GARCH t-Student VaR (99%) threshold (rolling 250-week window). Red dots mark the 13 violation weeks. Note the relatively even temporal distribution — consistent with the near-perfect Ch-P duration test result (p = 0.948) — in contrast to the violation clusters seen for EWMA at 99%.

3.3.4 All 14 Specifications Comparison

Table 3.7 provides the overview comparison across all models and tests, allowing patterns to be identified briefly.

Model	Level	Kupiec	Ch2	Ch-P	ES test	Overall
Normal (uncond.)	95%	✗	✗	✗	✗	FAIL
t-Student (uncond.)	95%	✗	✗	✗	✗	FAIL
Historical (uncond.)	95%	✗	✗	✗	✓	PARTIAL
Normal (uncond.)	99%	✓	✓	✗	✗	PARTIAL
t-Student (uncond.)	99%	✓	✓	✗	✗	PARTIAL
Historical (uncond.)	99%	✓	✗	✗	✓	PARTIAL
EWMA Normal	95%	✓	✓	✓	✗	PARTIAL
EWMA Normal	99%	✗	✗	✗	✗	FAIL
EWMA t-Student	95%	✓	✓	✓	✗	PARTIAL

EWMA t-Student	99%	✗	✗	✗	✗	FAIL
GARCH Normal	95%	✗	✗	✓	✓	PARTIAL
GARCH Normal	99%	✗	✗	✓	✗	PARTIAL
GARCH t-Student	95%	✗	✗	✓	✓	PARTIAL
GARCH t-Student	99%	✓	✓	✓	✓	PASS ⭑

Table 3.7: Overview comparison — all models, all tests, both confidence levels

Three cross-cutting patterns emerge from Table 3.7. First, **the ES test is the hardest to pass** it rejects for every model at the 95% level except Historical Simulation and GARCH t-Student, and at the 99% level only GARCH t-Student passes. This confirms the centrality of the distributional assumption for ES adequacy. Second, **GARCH consistently passes the Ch-P duration test** regardless of distributional assumption, while EWMA at 99% fails it. This reflects GARCH's reversion to equilibrium variance: even when the violation count is approximately correct, the variance returns toward its long-run level between shocks, preventing temporal clustering. Third, **no single model dominates at both confidence levels**: EWMA is better calibrated at 95% while GARCH t-Student is the only adequate specification at 99%. This implies that model selection should be confidence-level-specific, a practical recommendation that is rarely discussed in the existing literature.

3.4 Stress Test Analysis Across Market Regimes

The aggregate results in Section 3.3 mask important heterogeneity across market environments. The out-of-sample period is disaggregated into four regimes with distinct volatility characteristics: **Calm 2013–2019** (365 observations, VIX average ~14), **COVID-19 2020** (52 observations, VIX peak ~66), **Rate Hike 2022** (52 observations, sustained moderate volatility over ~40 weeks), and **Post-crisis 2023–2026** (169 observations, stabilization phase). The target violation rates are 5% and 1% at the 95% and 99% levels respectively.

3.4.1 Unconditional Models

Period	Obs.	Normal	t-Student	Historical	Assessment
Calm 2013–2019	365	1.1%	0.8%	0.3%	All far too conservative
COVID 2020	52	7.7%	5.8%	5.8%	Normal fails; others borderline
Rate Hike 2022	52	3.8%	0.0%	0.0%	t-Student & Historical over-protect

Post-2022	169	0.6%	0.6%	0.6%	All too conservative
Full Sample	730	1.5%	1.0%	0.7%	All undershoot the 5% target

Table 3.8: Stress test results — unconditional models, violation rates by regime (95% VaR, target: 5%)

The Calm 2013–2019 period reveals the most extreme failure of unconditional models: violation rates of 0.3–1.1% against a 5% target represent a 78-94% shortfall. Full-sample variance estimates anchored to the GFC and COVID episodes massively overstate risk during the calm period, imposing unnecessary capital costs on institutions using static models. COVID-19 produces violation rates of 5.8–7.7% at 95% - above target, consistent with the impossibility of anticipating a black swan event from historical data. During the Rate Hike 2022 period, t-Student and Historical models produce zero violations, indicating over-protection under a sustained volatility regime distinct from the crisis episodes dominating the full-sample estimates. The notable full-sample result is the t-Student model's near-perfect calibration at 99% (violation rate 1.0% against target 1%), suggesting that the fat-tailed parametric distribution captures the unconditional deep-tail loss frequency correctly over a full sample including crisis events.

3.4.2 Dynamic Models

Period	Obs.	EWMA Norm	EWMA t-Std	GARCH Norm	GARCH t-Std	Assessment
Calm 2013–2019	365	4.9%	5.5%	6.3%	6.8%	All near target — huge improvement
COVID 2020	52	7.7%	7.7%	7.7%	9.6%	All fail — black swan
Rate Hike 2022	52	5.8%	5.8%	7.7%	7.7%	EWMA better than GARCH
Post-2022	169	3.6%	3.6%	6.5%	7.1%	All acceptable
Full Sample	730	4.4%	4.7%	6.7%	7.3%	EWMA best calibrated at 95%

Table 3.9: Stress test results — dynamic models, violation rates by regime (95% VaR, target: 5%)

Period	Obs.	EWMA Norm	EWMA t-Std	GARCH Norm	GARCH t-Std	Assessment
Calm 2013–2019	365	2.5%	2.2%	2.7%	2.2%	All too many violations
COVID 2020	52	5.8%	5.8%	3.8%	3.8%	GARCH better than EWMA

Rate Hike 2022	52	1.9%	1.9%	1.9%	1.9%	All identical — borderline
Post-2022	169	1.2%	1.2%	1.2%	1.2%	All perform identically well
Full Sample	730	2.1%	1.9%	2.1%	1.8%	GARCH t-Std best (1.8%)

Table 3.10: Stress test results — dynamic models, violation rates by regime (99% VaR, target: 1%)

Calm 2013–2019. Dynamic models achieve 4.9–6.8% at 95% - all within a plausible range of the 5% target, compared with 0.3–1.1% for unconditional models. This is the most important finding of the stress test analysis and strongly confirms the first research hypothesis. EWMA Normal achieves the closest match to the target (4.9%), slightly undercounting relative to GARCH's slight over-count. At 99%, however, dynamic models substantially overshoot: violation rates of 2.2–2.7% are two to three times the 1% target. The calm period catches the dynamic models in the opposite direction at the strictest confidence level - reactive in low-volatility conditions, they generate excess false alarms at the 1% threshold. This pattern is consistent with the findings of Christoffersen (1998), who notes that GARCH tends to over-fit short-horizon variance spikes during calm periods.

COVID-19 crash (2020). All models fail at both confidence levels. At 95%, violation rates of 7.7–9.6% are well above the 5% target across the board; at 99%, GARCH produces 3.8% versus EWMA's 5.8%. The marginal GARCH outperformance at 99% reflects its stronger initial variance response to the first large weekly shock of the COVID episode, which partially offsets the subsequent spike. However, the fundamental conclusion is unavoidable: no backward-looking statistical model can adequately anticipate a black swan event. This is a limitation inherent to the entire class of historical volatility models, not a failure specific to any specification. The finding is consistent with Taleb (2007)'s critique of VaR and with the empirical documentation in McNeil et al. (2015).

Rate Hike 2022. This period reveals the most consequential and previously undocumented distinction between EWMA and GARCH in this thesis. At the 95% level, EWMA produces violation rates of 5.8% - close to the 5% target, while GARCH produces 7.7%. The 2022 stress episode was a sustained, gradual bear market: rather than a single catastrophic shock, it consisted of roughly 40 weeks of moderate weekly losses spread over the full calendar year. GARCH responds strongly to each individual weekly loss given its high ARCH coefficient ($\alpha = 0.32$), continuously re-estimating the conditional variance upward and pushing the VaR threshold lower. As the market continues falling in subsequent weeks, these newly lowered thresholds are violated - generating an excess of false alarms. EWMA's smoother exponential weighting absorbs these successive shocks more gradually and tracks the sustained stress level more accurately. This distinction is not apparent in the aggregate backtesting results of Section 3.3, demonstrating the value of regime-disaggregated analysis for understanding model behaviors.

3.5 Discussion

The results across all 14 model–distribution combinations, four backtesting tests at two confidence levels, and four distinct market regimes yield a coherent set of conclusions about when each model works and why.

First: dynamic models substantially outperform unconditional models at 95%. Rolling-window estimation reduces the under-count from 60%+ (unconditional models: 0.3–1.1%) to near-target (dynamic models: 4.9–6.8%) during the calm 2013–2019 period. This is the finding that most clearly validates the practical value of conditional volatility models. Institutions using static models during extended calm periods systematically over-hold regulatory capital. This result aligns with the theoretical argument of Engle and Manganelli (2004) that dynamic quantile models are needed to track time-varying conditional distributions.

Second: the t-Student distribution is the binding constraint for ES accuracy at 99%. GARCH Normal and GARCH t-Student produce nearly identical VaR violation counts at 99%, but GARCH Normal fails the McNeil-Frey ES test ($p = 0.047$) while GARCH t-Student passes ($p = 0.154$). The difference is entirely attributable to the ES multiplier: the t-Student with $\nu = 9.17$ assigns a 20% larger ES per unit of VaR at the 99% level (Table 2.1). This confirms that getting the distributional assumption right is not merely a theoretical concern, it is the difference between passing and failing a formal regulatory backtest. This finding is consistent with the results of McNeil and Frey (2000), who emphasized the importance of fat-tailed innovation distributions for ES estimation in heteroskedastic models.

Third: GARCH t-Student is the only specification meeting all regulatory requirements at 99%. The combination of equilibrium-reverting GARCH dynamics and fat-tailed t-Student innovations is jointly necessary: neither component alone is sufficient. EWMA t-Student fails because its fixed decay parameter generates too many violations under sudden volatility spikes (violation clustering → Ch2 failure). GARCH Normal fails because its thin-tailed distribution underestimates ES even when the violation count is approximately correct. Only GARCH t-Student simultaneously achieves violation count adequacy (Kupiec $p = 0.069$), temporal independence (Ch2 $p = 0.089$, Ch-P $p = 0.948$), and ES calibration (ES $p = 0.154$). For institutions required to demonstrate model adequacy under Basel III internal models' approach, this is the only specification in this study that would pass.

Fourth: EWMA is preferable for 95% VaR and for sustained stress episodes. At 95%, EWMA Normal achieves the best full-sample calibration (4.4%) and handles the 2022 rate shock more gracefully than GARCH (5.8% vs. 7.7% violation rate). EWMA's fixed decay structure, while theoretically inferior to GARCH's maximum-likelihood mean-reversion, proves more stable in exactly the type of moderate, sustained stress that characterized the 2022 bear market. The practical implication is that model selection should be purpose-specific: GARCH t-Student for Basel III 99% ES reporting; EWMA for internal 95% VaR monitoring.

Fifth: no model is robust to COVID-19. The March 2020 crash defeats every model at both confidence levels. This is not a finding specific to the models tested - it is an inherent limitation of the class of backward-looking statistical risk models. No amount of sophistication in the volatility dynamics or distributional assumptions provides protection against an event with no historical precedent in the estimation window. The practical implication is that scenario-based stress testing against tail events - rather than model-based VaR and ES alone - is essential for comprehensive risk management. This conclusion is consistent with the post-GFC emphasis on stress testing in the Basel III framework (BCBS, 2009) and with the extensive literature on model limitations documented by Taleb (2007).

SUMMARY

This thesis compares the Exponentially Weighted Moving Average (EWMA) model and the GARCH (1,1) model - each estimated under Normal and t-Student distributional assumptions -for measuring and backtesting Value at Risk (VaR) and Expected Shortfall (ES) for the S&P 500 index. The empirical analysis is based on approximately 1,000 weekly log-returns from January 2007 to March 2026, with out-of-sample backtesting conducted over approximately 730 weeks using a rolling 250-week estimation window. Four formal backtesting procedures are applied to all 14 model–distribution combinations: the Kupiec (1995) unconditional coverage test, the Christoffersen (1998) independence test, the Christoffersen and Pelletier (2004) duration test, and the McNeil and Frey (2000) Expected Shortfall test. A model is assessed as PASS only if all four tests simultaneously fail to reject at the 5% significance level.

Five conclusions emerge from the analysis. Rolling-window conditional volatility models prove substantially more accurate than their unconditional counterparts when market conditions are calm: during 2013–2019, EWMA and GARCH achieve violation rates of 4.9–6.8% at the 95% level, whereas static models fall to 0.3–1.1% — a failure attributable to full-sample variance estimates dominated by crisis-era observations. The choice of distributional tail shape is the decisive factor for ES accuracy at 99%: although GARCH Normal and GARCH t-Student record near-identical violation frequencies, only the fat-tailed specification passes the McNeil–Frey test ($p = 0.154$ vs. $p = 0.047$), with the difference traced directly to the t-Student's larger ES multiplier at $\nu = 9.17$. Of all fourteen model–distribution combinations evaluated, GARCH (1,1) with t-Student innovations is the sole specification to satisfy all four regulatory criteria concurrently at the 99% confidence level, and is therefore recommended for Basel III internal-models reporting. At the 95% level the preference reverses: EWMA achieves a closer match to the target violation rate (4.4%) and, unlike GARCH, avoids over-reacting to the sequence of moderate weekly losses that characterised the 2022 rate-hike cycle. Finally, backward-looking statistical models as a class — regardless of specification — cannot provide adequate protection against an event with no historical precedent in the estimation window, as demonstrated by the uniform failure across all models during the March 2020 crash.**Fifth**, no model adequately handles the COVID-19 crash of March 2020, an inherent limitation of all backward-looking statistical risk models rather than a failure of any specification.

The main limitations of this study are the following. The analysis is restricted to a single asset class (US large-cap equity) and a single holding period (one week), and results may not generalize directly to other markets or frequencies. The symmetric GARCH (1,1) specification does not capture the leverage effect documented by Black (1976), and a more complete analysis would include EGARCH or IGARCH as a robustness check. The rolling window length of 250 weeks is a methodological choice; results may be sensitive to this parameter, particularly in the stress test analysis. Finally, the analysis is confined to a single decay parameter ($\lambda = 0.94$) for EWMA; a sensitivity analysis over alternative values of λ would strengthen the robustness of the conclusions.

Directions for future research include three natural extensions. First, incorporating an asymmetric model EGARCH or IGARCH to test whether the leverage effect materially improves 99% ES calibration. Second, applying the framework to other asset classes and geographies to assess the generalizability of the main findings. Third, investigating the Stochastic Volatility (SV) model of Kim, Shephard and Chib (1998) as an alternative to GARCH: SV models offer a richer latent-variance structure but require Bayesian estimation, making their practical backtesting performance an open empirical question.

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